MATH120-24S2 Discrete Mathematics

Assignment 7

Due: 11:59pm Friday 20 September
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1. Let A, B and C be subsets of a universal set U. Prove or disprove the following statements.

   (a) If $A \cap B = \emptyset$ then $A = \emptyset$ or $B = \emptyset$.

   (b) If $A \subseteq B$, then $A \cup C \subseteq B \cup C$.

   Solution:

   (a) False. For example, if $A = \{1\}$ and $B = \{2\}$, then $A \cap B = \emptyset$ but neither A nor B is the emptyset.

   (b) True.
   Suppose $A \subseteq B$ and let $x \in A \cup C$. We will show that $x \in B \cup C$. Since $x \in A \cup C$, we have that $x \in A$ or $x \in C$. However, $A \subseteq B$ so if $x \in A$ then $x \in B$. Therefore $x \in B$ or $x \in C$ so $x \in B \cup C$. Therefore $A \cup C \subseteq B \cup C$.

2. *Hint: To show $X = Y$, you need to show both $X \subseteq Y$ and $Y \subseteq X$.*
   Let A, B, and C be subsets of a universal set.

   (a) Prove that

   $$(A \cup B) - C = (A - C) \cup (B - C).$$

   (b) We asked ChatGPT to prove that if $A \subseteq B$ then $A \cap C = B \cap C$. It correctly proved that $A \cap C \subseteq B \cap C$, but then continued with:
   To prove: $B \cap C \subseteq A \cap C$.
   Assumption: $A \subseteq B$.
   Let $x \in B \cap C$. By definition of intersection, $x \in B$ and $x \in C$. Since $A \subseteq B$, if $x \in B$ and $x \in A \subseteq B$, $x \in A$. Therefore $x \in A$

and $x \in C$, so $x \in A \cap C$. Hence, $x \in B \cap C$ implies $x \in A \cap C$. Thus, $B \cap C \subseteq A \cap C$.

Conclusion: We have shown that $A \cap C = B \cap C$.

What is the mistake in this part of the proof? Provide a counterexample or a correct proof of the statement: "If $A \subseteq B$, then $B \cap C \subseteq A \cap C$."

Solution:

(a) We will show that $(A \cup B) - C \subseteq (A - C) \cup (B - C)$ and also that $(A - C) \cup (B - C) \subseteq (A \cup B) - C$.

Let $x \in (A \cup B) - C$, then x is in A or B but $x \notin C$. Hence x is in $A - C$ or $x \in B - C$. Therefore $x \in (A - C) \cup (B - C)$ and we have shown that $(A \cup B) - C \subseteq (A - C) \cup (B - C)$.

Next, let $x \in (A - C) \cup (B - C)$. Then $x \in A - C$ or $x \in B - C$. If $x \in A - C$ then $x \in A$ and $x \notin C$. Similarly, if $x \in B - C$ then $x \in B$ and $x \notin C$. In either case we have x in A or B but $x \notin C$ and hence $x \in (A \cup B) - C$.

(b) The second part of this proof starts with an assumption that $A \subseteq B$ (which is a correct assumption). However in the second line below that, it states "Since $A \subseteq B$, **if** $x \in B$... **then** $x \in A$." This is not a logical implication since we could take $x \in B$ so that x is **not** in A. The remainder of the argument relied on the fact that $x \in A$ which is not a valid assumption.

A counter-example to the statement is $A = \{1\}$, $B = \{1, 2, 3\}$ and $C = \{1, 2, 4\}$. Then $B \cap C = \{1, 2\}$ and $A \cap C = \{1\}$. Then we have that $A \subseteq B$ but $B \cap C \not\subseteq A \cap C$

3. Consider the relation $\mathcal{R}$ on $\mathbb{Z}$ given by

$$\mathcal{R} = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} : \ 3 \mid (a^2 - b^2)\}.$$

(a) List three elements in the relation and three elements that are in $\mathbb{Z} \times \mathbb{Z}$ but not in $\mathcal{R}$.

(b) Show that $\mathcal{R}$ is an equivalence relation.

(c) Describe the distinct equivalence classes of $\mathcal{R}$.

Solution:

(a) E.g. $(1,1),(6,3),(5,2),(1,2),(4,5) \in \mathcal{R}$ and $(0,2),(3,5),(1,3) \notin \mathcal{R}$

(b) To show that $\mathcal{R}$ is an equivalence relation we must show that it is reflexive, symmetric and transitive.
Reflexive: Let $m \in \mathbb{Z}$. Since $m^2 - m^2 =$, we have $4 \mid m^2 - m^2$ and hence $(m,m) \in \mathcal{R}$. Therefore $\mathcal{R}$ is reflexive. **Symmetric**: Suppose $(m,n) \in \mathcal{R}$ for some $m,n \in \mathbb{Z}$. Then $3 \mid m^2 - n^2$. Since $n^2 - m^2 = -(m^2 - n^2)$, it follows that $3 \mid n^2 - m^2$ and hence that $(n,m) \in \mathcal{R}$. Therefore $\mathcal{R}$ is symmetric. **Transitive**: Suppose $(\ell,m),(m,n) \in \mathcal{R}$ for some $\ell,m,n \in \mathbb{Z}$. Then $3 \mid \ell^2 - m^2$ and $3 \mid m^2 - n^2$. By linearity of divisibility, it follows that $3 \mid (\ell^2 - m^2) + (m^2 - n^2)$ and hence $3 \mid \ell^2 - n^2$. Therefore $(\ell,n) \in \mathcal{R}$ and hence $\mathcal{R}$ is transitive.

(c) The equivalence classes of $\mathcal{R}$ are:

$$[0] = \{3n : n \in \mathbb{Z}\} \text{ and}$$
$$[1] = [2] = \{3n+1, 3n+2 : n \in \mathbb{Z}\}.$$

To see that $[0] = \{3n : n \in \mathbb{Z}\}$ is an equivalence class, we check that $3 \mid m^2 - n^2$ for any m and n such that $m \equiv n \equiv 0 \mod 3$. Let $m = 3a$ and $n = 3b$ for some $a,b \in \mathbb{Z}$, then $m^2 - n^2 = 9a^2 - 9b^2 = 9(a^2 - b^2)$, and hence $m^2 - n^2$ is divisible by 3 because $a^2 - b^2$ is an integer. To see that $[1] = \{3n+1, 3n+2 : n \in \mathbb{Z}\}$ is an equivalence class, we first check that $3 \mid (3m+1)^2 - (3n+1)^2$ which holds since $(3m+1)^2 - (3n+1)^2 = 9m^2 + 6m - 9n^2 - 6n$. We also check that $3 \mid (3m+2)^2 - (3n+2)^2$ which holds since $(3m+2)^2 - (3n+2)^2 = 9m^2 + 12m - 9n^2 - 12n$. Finally, we check that $3 \mid (3m+1)^2 - (3n+2)^2$ which holds since $(3m+1)^2 - (3n+2)^2 = 9m^2 + 6m + 1 - 9n^2 - 12n - 4$.